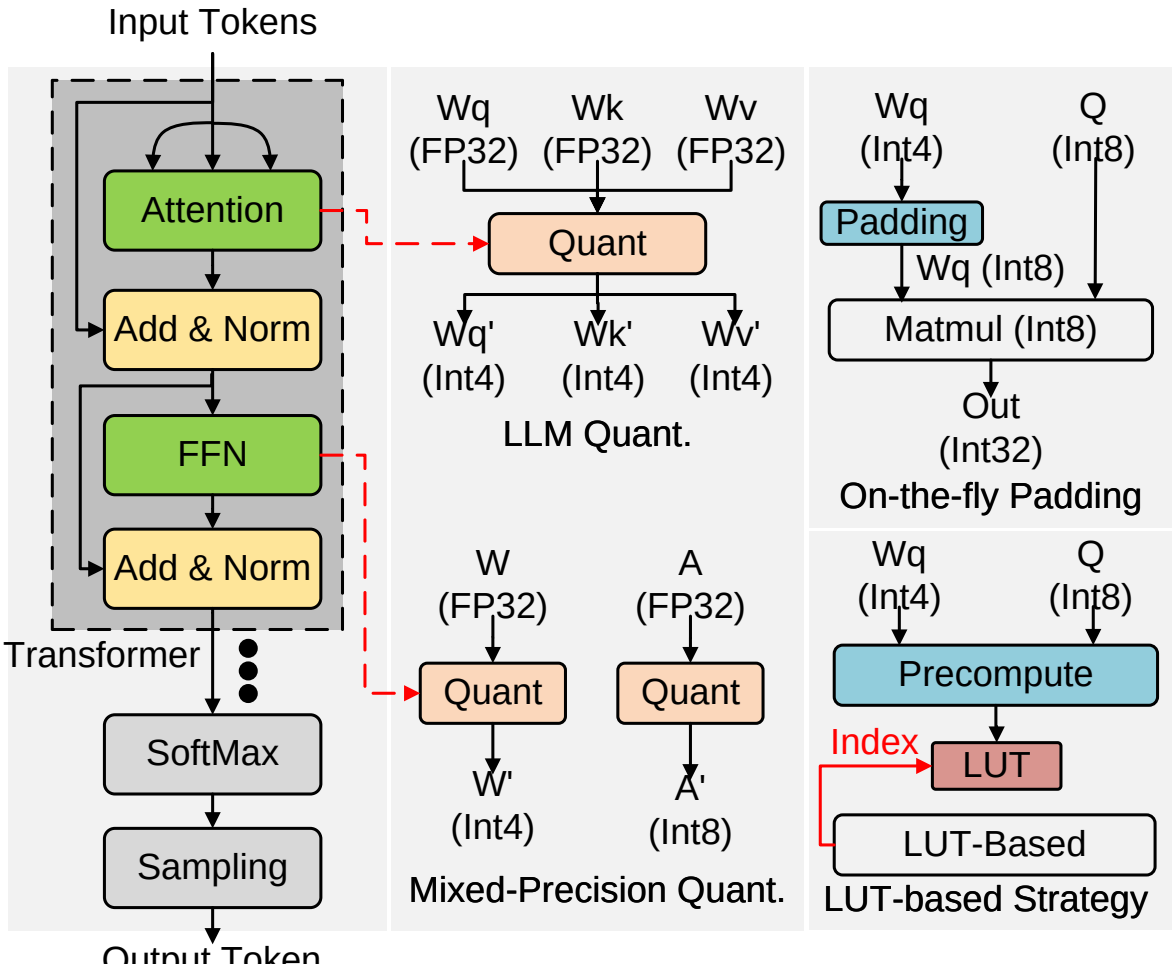
Input Tokens
Attention
Add & Norm
FFN
Add & Norm
Transformer
SoftMax
Sampling
Output Token
Wq (FP32)
Wk (FP32)
Wv (FP32)
Quant
Wq' (Int4)
Wk' (Int4)
Wv' (Int4)
LLM Quant.
Wq (Int4)
Q (Int8)
Padding
Wq (Int8)
Matmul (Int8)
Out (Int32)
On-the-fly Padding
W (FP32)
A (FP32)
Quant
Quant
W' (Int4)
A' (Int8)
Mixed-Precision Quant.
Wq (Int4)
Q (Int8)
Precompute
Index
LUT
LUT-Based
LUT-based Strategy